Power Budget EGR 304

Team Number:	209
Project Name:	**Automated Laundry Folding Device**
Team Member Names:	**Kristen Giovanetto, Ruben Esquer, Diego Suarez, Zainan Zarif**
Version:	**1**

A. List ALL major components (active devices, integrated circuits, etc.) except for power sources, voltage regulators,

All Major Components	Component Name	Part Number	Supply	#	Absolute	Total	Unit
	PIC18F57Q43 Curiosity Nanc	DM164150	1.8 - 5.5 V	1	100	100	mA
	Op-Amp	MCP6004-I/P-NI	1.8 - 6.0 V	1	0.68	0.68	mA
	Photoresistor	PDV-P8103-ND	(150 V absolu	1		0	mA
	5V Voltage Regulator L78(	497-15681-5-	7.5 - 18 V	1	6	6	mA
						0	mA

B. Assign each major component above to ONE power rail below. Try to minimize the number of different power rails in

+12V Power Rail	Component Name	Part Number	Supply	#	aximumCurre	alCurrent(mA)	Unit
						0	mA
						0	mA
						0	mA
						0	mA
						0	mA
					Subtotal	0	mA
					Safety Margin	25%	
					Total Current Required on +12V Rail	0	mA
c1. Regulator or Source Choice						0	mA
					Total Remaining Current Available on +12V Rail	0	mA

+5V Power Rail	Component Name	Part Number	Supply	#	Absolute	alCurrent(mA)	Unit
	PIC18F57Q43 Curiosity Nanc	DM164150	1.8 - 5.5 V	1	100	100	mA
	Opamp	MCP6004-I/P-N	1.8 - 6.0 V	1	0.68	0.68	mA
	Photoresistor	PDV-P8103-ND	Not specified	1		0	mA
						0	mA
						0	mA
					Subtotal	100.68	mA
					Safety Margin	25%	
					Total Current Required on +5V Rail	125.85	mA
c2. Regulator or Source Ch	+5V Regulator	L497-15681-5-	7.5 - 18 V	1	6	6	mA
					Total Remaining Current Available on +5V Rail	-119.85	mA

-5V Power Rail	Component Name	Part Number	yVoltageRar	#	aximumCurre	alCurrent(mA)	Unit
							mA
						0	mA
						0	mA
						0	mA
					Subtotal	0	mA
					Safety Margin	25%	
					Total Current Required on -5V Rail	0	mA
c3. Regulator or Source Ch						0	mA
					Total Remaining Current Available on -5V Rail	0	mA

+3.3V Power Rail	Component Name	Part Number	yVoltageRar	#	aximumCurre	alCurrent(mA)	Unit
							mA
						0	mA
						0	mA
						0	mA
					Subtotal	0	mA
					Safety Margin	25%	
					Total Current Required on +3.3V Rail	0	mA
c4. Regulator or Source Ch						0	mA
					Total Remaining Current Available on 3.3V Rail	0	mA

C. For each power rail above, select a specific voltage regulator using the same process as for major component selection.

D. Select a specific external power source (wall supply or battery) for your system, and confirm that it can supply all of the

External Power Source 1	Component Name	Part Number	Supply	Outpu	Absolute	Total	Unit
Power Source 1 Selection	Plug-in Wall Supply	1939-1456-ND	90–264 V AC	9 V DC	1000	1000	mA
						0	mA
Power Rails Connected to External Power Source 1	+5V Regulator	497-15681-5-N	7.5 - 18V	5	6	30	mA
						0	mA
					Total Remaining Current Available on External Power Source 1	970	mA

External Power Source 2	Component Name	Part Number	yVoltageRar	Outpu	aximumCurre	alCurrent(mA)	Unit
Power Source 2 Selection							mA
Power Rails Connected to External Power Source 2						0	mA
					Total Remaining Current Available on External Power Source 2	0	mA

E. Calculate Battery Life (if applicable). For each battery, also check the worst-case lifetime of the

Component Name	Part Number	yVoltageRange	Capacity(mAh)	redByRegulators	
				0	
			Battery Life	#DIV/0!	hours

Notes

External Supply Voltage should be determined by the dropout voltage for highest-voltage regulator (e.g., +14V for a +12V regulator).
If you have multiple units in your design (e.g., a base unit and remote unit) then you need a separate power budget for each unit